

Sr. No.	Error	Error Line Number	Error Type	Solution
1	<code>int marks[5] = {80,90,70,60,50}</code>	6	Syntax Error	Missing semicolon after array initializer. Add ; at the end of the line.
2	<code>for(int i=0;i<=5;i++) { total += marks[i]; }</code>	12	Runtime Error	Array index out of bounds; marks[5] is invalid for a 5-element array. Use i<5.
3	<code>average = total / 5;</code>	21	Logical Error	Integer division truncates the result. Use average = (float)total / 5;
4	<code>printf("Average = %d\n", average);</code>	23	Semantic Error	Wrong format specifier '%d' used for a float. Use printf("Average = %f\n", average);
5	<code>scores[5] = 60;</code>	35	Runtime Error	Array index out of bounds; valid indices for scores[5] are 0 to 4.
6	<code>char subject[5]; strcpy(subject, "Programming");</code>	47	Runtime Error	Buffer overflow: "Programming" needs 12 bytes but subject only holds 5. Declare char subject[20];
7	<code>attendance["one"] = 1;</code>	55	Compile Error	Array subscript must be an integer, not a string. Use a valid index, e.g. attendance[0] = 1;
8	<code>attendance[2.5] = 1;</code>	57	Compile Error	Array subscript must be an integer, not a float. Use attendance[2] = 1;
9	<code>for(int i=0;i<=5;i++) { printf("%d", attendance[i]); }</code>	65	Runtime Error	Array index out of bounds; attendance[5] is invalid. Use i<5.
10	<code>scanf("%d", arr[i]);</code>	77	Compile Error	scanf needs a pointer, not a value. Use scanf("%d", &arr[i]);
11	<code>grade[0] = "A";</code>	93	Compile Error	Cannot assign a string literal to a char. Use grade[0] = 'A'; (single quotes).
12	<code>if(prices[2] = 300)</code>	102	Logical Error	Assignment '=' used instead of comparison '=='. Use if(prices[2] == 300)
13	<code>inventory = prices;</code>	111	Compile Error	Arrays cannot be assigned directly. Copy elements individually or use memcpy().
14	<code>printf("%d\n", salary[0]);</code>	121	Semantic Error	Wrong format specifier '%d' used for a float. Use printf("%f\n", salary[0]);
15	<code>city = "Jaipur";</code>	127	Compile Error	Arrays cannot be assigned directly. Use strcpy(city, "Jaipur");
16	<code>for(int i=0;i<=5;i++) { result[i] = i * 10; }</code>	134	Runtime Error	Array index out of bounds; result[5] is invalid. Use i<5.

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1	<code>for(int i=0;i<=10;i++)</code> used to print numbers[] (array size is 10, valid indices 0-9)	17	Runtime Error (Array Out of Bounds)	Change loop condition to <code>i<10</code> so it stays within valid indices.
2	<code>sum = numbers[i];</code> overwrites sum instead of accumulating it	28	Logical Error	Use <code>sum += numbers[i];</code> to accumulate the total instead of overwriting it.
3	<code>average = sum / 10;</code> performs integer division, losing decimal part	35	Logical Error	Use <code>average = sum / 10.0f;</code> (or cast sum to float) to get a correct decimal average.
4	<code>printf("Average = %d\n", average);</code> uses %d for a float variable	37	Format Specifier Error	Use %f instead of %d, e.g. <code>printf("Average = %.2f\n", average);</code>
5	<code>if(marks[i] < highest)</code> finds the lowest mark instead of the highest	49	Logical Error	Change the comparison to <code>if(marks[i] > highest)</code> to correctly find the highest mark.
6	<code>attendance[5] = 1;</code> writes outside the array (size 5, valid indices 0-4)	67	Runtime Error (Array Out of Bounds)	Increase array size to <code>attendance[6]</code> or remove this out-of-range assignment.
7	<code>for(int i=0;i<=5;i++)</code> reads <code>attendance[5]</code> , which is out of bounds	71	Runtime Error (Array Out of Bounds)	Change loop condition to <code>i<5</code> so only valid indices are accessed.
8	<code>strcpy(subject,"ComputerScience");</code> string (15 chars+null) is larger than <code>subject[10]</code>	82	Runtime Error (Buffer Overflow)	Increase array size, e.g. <code>char subject[20];</code> or use a shorter string.
9	<code>branch = "IT";</code> tries to assign a string literal directly to an array name	90	Compile-Time Error (Invalid Assignment)	Use <code>strcpy(branch, "IT");</code> to copy the string into the array instead of assigning.
10	<code>printf("%s\n", college);</code> <code>college[]</code> is filled char by char but never null-terminated	104	Runtime Error (Undefined Behavior / Missing Null Terminator)	Add <code>college[5] = '\0';</code> after assigning the characters, and size the array as <code>college[6]</code> or larger.
11	<code>fees["two"] = 50000;</code> uses a string literal as an array index	110	Compile-Time Error (Invalid Array Subscript)	Use a valid integer index, e.g. <code>fees[1] = 50000;</code>
12	<code>fees[3.5] = 60000;</code> uses a non-integer (float) array index	112	Compile-Time Warning/Error (Invalid Array Subscript)	Use an integer index, e.g. <code>fees[3] = 60000;</code>

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13	scanf("%d", result[i]); passes the value instead of the address of the element	132	Compile-Time Warning / Runtime Error (Wrong Argument Type to scanf)	Use scanf("%d", &result[i]); to pass the address of the element.
14	if(stock[2] = 300) uses assignment (=) instead of comparison (==)	150	Logical Error (Assignment instead of Comparison)	Use if(stock[2] == 300) to correctly compare the values.
15	backup = stock; tries to assign one array directly to another	159	Compile-Time Error (Invalid Array Assignment)	Copy elements individually in a loop, or use memcpy(backup, stock, sizeof(stock));
16	printf("%d\n", salary[0]); uses %d for a float array element	169	Format Specifier Error	Use %f instead of %d, e.g. printf("%.2f\n", salary[0]);
17	for(int i=0;i<=5;i++) writes inventory[5], which is out of bounds (size 5, valid 0-4)	175	Runtime Error (Array Out of Bounds)	Change loop condition to i<5 so only valid indices are accessed.
18	strcpy(city,"JaipurCity"); string (10 chars+null) is larger than city[8]	191	Runtime Error (Buffer Overflow)	Increase array size, e.g. char city[20]; or use a shorter string.
19	count = inventory; tries to assign one array directly to another	199	Compile-Time Error (Invalid Array Assignment)	Copy elements individually in a loop, or use memcpy(count, inventory, sizeof(inventory));
20	printf("Program Ended\n") is missing a terminating semicolon	205	Compile-Time Error (Syntax Error - Missing Semicolon)	Add a semicolon at the end: printf("Program Ended\n");

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1	average = total / 5; performs integer division, losing the decimal part	21	Logical Error	Use average = total / 5.0f; (or cast total to float) to get a correct decimal average.
2	if(marks[i] > lowest) finds the highest mark again instead of the lowest	45	Logical Error	Change the comparison to if(marks[i] < lowest) to correctly find the lowest mark.
3	for(int i=0;i<=5;i++) reads attendance[5], which is out of bounds (size 5, valid 0-4)	65	Runtime Error (Array Out of Bounds)	Change loop condition to i<5 so only valid indices are accessed.
4	strcpy(subject,"Programming"); string (11 chars+null) is larger than subject[8]	76	Runtime Error (Buffer Overflow)	Increase array size, e.g. char subject[15]; or use a shorter string.
5	scanf("%s",&branch); passes the address of the whole array (wrong pointer type) instead of the array itself	84	Compile-Time Warning (Incompatible Pointer Type)	Use scanf("%s", branch); since an array name already decays to a pointer to its first element.
6	fees[5] = 100000; writes outside the array (size 5, valid indices 0-4)	98	Runtime Error (Array Out of Bounds)	Increase array size to fees[6] or remove this out-of-range assignment.
7	if(stock[2] = 30) uses assignment (=) instead of comparison (==)	111	Logical Error (Assignment instead of Comparison)	Use if(stock[2] == 30) to correctly compare the values.
8	backup = stock; tries to assign one array directly to another	120	Compile-Time Error (Invalid Array Assignment)	Copy elements individually in a loop, or use memcpy(backup, stock, sizeof(stock));
9	printf("%d\n", salary[0]); uses %d for a float array element	130	Format Specifier Error	Use %f instead of %d, e.g. printf("%.2f\n", salary[0]);
10	scanf("%d", result[i]); passes the value instead of the address of the element	138	Compile-Time Warning / Runtime Error (Wrong Argument Type to scanf)	Use scanf("%d", &result[i]); to pass the address of the element.
11	city = "Jaipur"; tries to assign a string literal directly to an array name	156	Compile-Time Error (Invalid Assignment)	Use strcpy(city, "Jaipur"); to copy the string into the array instead of assigning.
12	inventory["one"] = 100; uses a string literal as an array index	164	Compile-Time Error (Invalid Array Subscript)	Use a valid integer index, e.g. inventory[1] = 100;

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13	inventory[2.5] = 200; uses a non-integer (float) array index	166	Compile-Time Warning/Error (Invalid Array Subscript)	Use an integer index, e.g. inventory[2] = 200;
14	for(int i=0;i<=5;i++) reads inventory[5], which is out of bounds (size 5, valid 0-4)	175	Runtime Error (Array Out of Bounds)	Change loop condition to i<5 so only valid indices are accessed.
15	count = inventory; tries to assign one array directly to another	196	Compile-Time Error (Invalid Array Assignment)	Copy elements individually in a loop, or use memcpy(count, inventory, sizeof(inventory));
16	strcpy(course,"Computer"); string (8 chars+null) is larger than course[6]	204	Runtime Error (Buffer Overflow)	Increase array size, e.g. char course[10]; or use a shorter string.
17	for(int i=0;i<=5;i++) writes numbers[5], which is out of bounds (size 5, valid 0-4)	212	Runtime Error (Array Out of Bounds)	Change loop condition to i<5 so only valid indices are accessed.
18	return 0 is missing a terminating semicolon	228	Compile-Time Error (Syntax Error - Missing Semicolon)	Add a semicolon at the end: return 0;

Sr No.	Error	Error Line Number	Error Type	Solution
1	averageSales = totalSales / 5; performs integer division, losing the decimal part	21	Logical Error (Wrong Formula)	Use averageSales = totalSales / 5.0f; (or cast to float) to get a correct decimal average.
2	printf("Average Sales = %d\n", averageSales); uses %d for a float variable	23	Type Mismatch (Format Specifier Error)	Use %f instead of %d, e.g. printf("Average Sales = %.2f\n", averageSales);
3	if(sales[i] < highestSale) finds the lowest sale even though the variable is named highestSale	31	Logical Error (Wrong Condition)	Change the comparison to if(sales[i] > highestSale) to correctly find the highest sale.
4	monthlySales[5] = 250; writes outside the array (size 5, valid indices 0-4)	49	Array Bounds Error (Runtime)	Increase array size to monthlySales[6] or remove this out-of-range assignment.
5	for(int i=0;i<=5;i++) reads monthlySales[5], which is out of bounds	53	Array Bounds Error (Runtime)	Change loop condition to i<5 so only valid indices are accessed.
6	strcpy(product,"LaptopComputer"); string (14 chars+null) is larger than product[8]	62	Runtime Error (Buffer Overflow)	Increase array size, e.g. char product[20]; or use a shorter string.
7	category = "Electronics"; tries to assign a string literal directly to an array name	70	Type Mismatch (Invalid Assignment)	Use strcpy(category, "Electronics"); to copy the string into the array instead of assigning.
8	stock["first"] = 100; uses a string literal as an array index	78	Compile-Time Error (Invalid Array Subscript)	Use a valid integer index, e.g. stock[0] = 100;
9	stock[2.7] = 200; uses a non-integer (float) array index	80	Compile-Time Warning/Error (Invalid Array Subscript)	Use an integer index, e.g. stock[2] = 200;
10	scanf("%d", orders[i]); passes the value instead of the address of the element	100	Type Mismatch (Wrong Argument Type to scanf)	Use scanf("%d", &orders[i]); to pass the address of the element.
11	if(revenue[2] = 3000) uses assignment (=) instead of comparison (==)	118	Logical Error (Wrong Condition)	Use if(revenue[2] == 3000) to correctly compare the values.
12	backupRevenue = revenue; tries to assign one array directly to another	127	Compile-Time Error (Invalid Array Assignment)	Copy elements individually in a loop, or use memcpy(backupRevenue, revenue, sizeof(revenue));
13	printf("%d\n", commission[0]); uses %d for a float array element	137	Type Mismatch (Format Specifier Error)	Use %f instead of %d, e.g. printf("%.2f\n", commission[0]);

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14	city = "Jaipur"; tries to assign a string literal directly to an array name	143	Compile-Time Error (Invalid Assignment)	Use strcpy(city, "Jaipur"); to copy the string into the array instead of assigning.
15	printf("%s\n", branch); branch[] is filled with only 4 characters and never null-terminated	156	Runtime Error (Missing Null Terminator)	Add branch[4] = '\0'; (and size the array as branch[6] or larger to fit it).
16	for(int i=0;i<=5;i++) writes employees[5], which is out of bounds (size 5, valid 0-4)	162	Array Bounds Error (Runtime)	Change loop condition to i<5 so only valid indices are accessed.
17	strcpy(department,"Marketing"); string (9 chars+null) is larger than department[6]	178	Runtime Error (Buffer Overflow)	Increase array size, e.g. char department[12]; or use a shorter string.
18	expenses = sales; tries to assign one array directly to another	186	Compile-Time Error (Invalid Array Assignment)	Copy elements individually in a loop, or use memcpy(expenses, sales, sizeof(sales));
19	ratings[-1] = 5; uses a negative index, writing before the start of the array	194	Array Bounds Error (Runtime, Negative Index)	Use a valid non-negative index within 0 to 4, e.g. ratings[0] = 5;
20	printf("%s\n", feedback[0]); feedback[0] is an int, not a string, used with %s	208	Type Mismatch (Wrong Format Specifier)	Use %d instead of %s, e.g. printf("%d\n", feedback[0]);
21	printf("Report Generated Successfully\n") is missing a terminating semicolon	212	Syntax Error (Missing Semicolon)	Add a semicolon at the end: printf("Report Generated Successfully\n");

Sr No.	Error	Error Line Number	Error Type	Solution
1	average = total / 10; performs integer division, losing the decimal part	21	Logical Error (Wrong Formula)	Use average = total / 10.0f; (or cast to float) to get a correct decimal average.
2	printf("Average = %d\n", average); uses %d for a float variable	23	Type Mismatch (Format Specifier Error)	Use %f instead of %d, e.g. printf("Average = %.2f\n", average);
3	if(marks[i] < highest) finds the lowest mark even though the variable is named highest	31	Logical Error (Wrong Condition)	Change the comparison to if(marks[i] > highest) to correctly find the highest mark.
4	if(marks[i] > lowest) finds the highest mark even though the variable is named lowest	45	Logical Error (Wrong Condition)	Change the comparison to if(marks[i] < lowest) to correctly find the lowest mark.
5	attendance[10] = 0; writes outside the array (size 10, valid indices 0-9)	62	Array Bounds Error (Runtime)	Increase array size to attendance[11] or remove this out-of-range assignment.
6	for(int i=0;i<=10;i++) reads attendance[10], which is out of bounds	66	Array Bounds Error (Runtime)	Change loop condition to i<10 so only valid indices are accessed.
7	strcpy(subject,"DataStructures"); string (14 chars+null) is larger than subject[10]	77	Runtime Error (Buffer Overflow)	Increase array size, e.g. char subject[20]; or use a shorter string.
8	department = "InformationTechnology"; tries to assign a string literal directly to an array name	85	Type Mismatch (Invalid Assignment)	Use strcpy(department, "InformationTechnology"); to copy the string into the array instead of assigning.
9	fees["two"] = 50000; uses a string literal as an array index	93	Compile-Time Error (Invalid Array Subscript)	Use a valid integer index, e.g. fees[1] = 50000;
10	fees[4.5] = 70000; uses a non-integer (float) array index	95	Compile-Time Warning/Error (Invalid Array Subscript)	Use an integer index, e.g. fees[4] = 70000;
11	scanf("%d", result[i]); passes the value instead of the address of the element	115	Type Mismatch (Wrong Argument Type to scanf)	Use scanf("%d", &result[i]); to pass the address of the element.
12	if(inventory[2] = 30) uses assignment (=) instead of comparison (==)	133	Logical Error (Wrong Condition)	Use if(inventory[2] == 30) to correctly compare the values.
13	backup = inventory; tries to assign one array directly to another	142	Compile-Time Error (Invalid Array Assignment)	Copy elements individually in a loop, or use memcpy(backup, inventory, sizeof(inventory));

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14	printf("%d\n", salary[0]); uses %d for a float array element	152	Type Mismatch (Format Specifier Error)	Use %f instead of %d, e.g. printf("%.2f\n", salary[0]);
15	city = "Jaipur"; tries to assign a string literal directly to an array name	158	Compile-Time Error (Invalid Assignment)	Use strcpy(city, "Jaipur"); to copy the string into the array instead of assigning.
16	printf("%s\n", branch); branch[] is filled with only 4 characters and never null-terminated	171	Runtime Error (Missing Null Terminator)	Add branch[4] = '\0'; (and size the array as branch[6] or larger to fit it).
17	for(int i=0;i<=5;i++) writes stock[5], which is out of bounds (size 5, valid 0-4)	177	Array Bounds Error (Runtime)	Change loop condition to i<5 so only valid indices are accessed.
18	strcpy(course,"Programming"); string (11 chars+null) is larger than course[8]	193	Runtime Error (Buffer Overflow)	Increase array size, e.g. char course[15]; or use a shorter string.
19	ratings[-1] = 10; uses a negative index, writing before the start of the array	201	Array Bounds Error (Runtime, Negative Index)	Use a valid non-negative index within 0 to 4, e.g. ratings[0] = 10;
20	printf("%s\n", feedback[0]); feedback[0] is an int, not a string, used with %s	209	Type Mismatch (Wrong Format Specifier)	Use %d instead of %s, e.g. printf("%d\n", feedback[0]);
21	scores = marks; tries to assign one array directly to another	215	Compile-Time Error (Invalid Array Assignment)	Copy elements individually in a loop, or use memcpy(scores, marks, sizeof(scores));
22	printf("%d\n", temp[5]); temp[5] is out of bounds (size 5, valid indices 0-4)	228	Array Bounds Error (Runtime)	Use a valid index within 0 to 4, e.g. printf("%d\n", temp[4]);